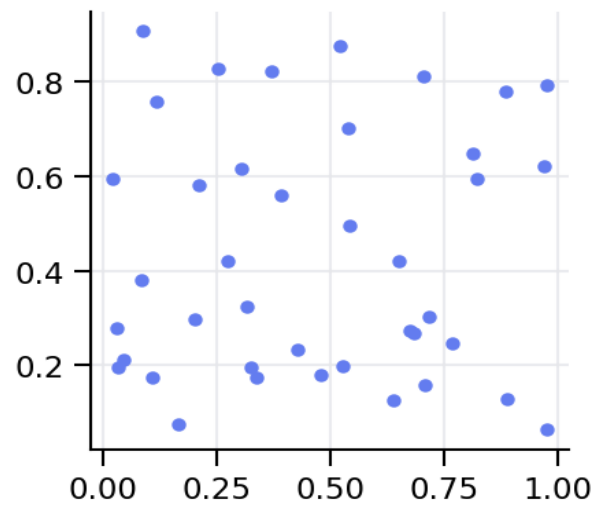


Clustering Algorithms – Detailed Study Notes

K-Means

K-Means clustering divides data into K groups by iteratively minimizing the distance between data points and cluster centroids. It is simple and fast but requires the number of clusters in advance.

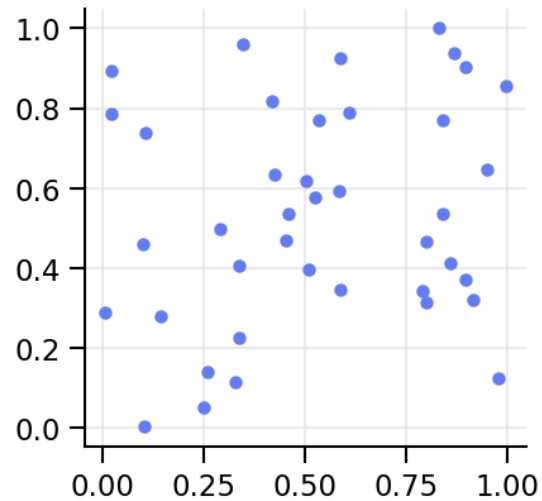
K-Means



Hierarchical Agglomerative

This algorithm builds clusters by progressively merging the closest clusters. It creates a dendrogram that helps visualize cluster relationships but is computationally expensive.

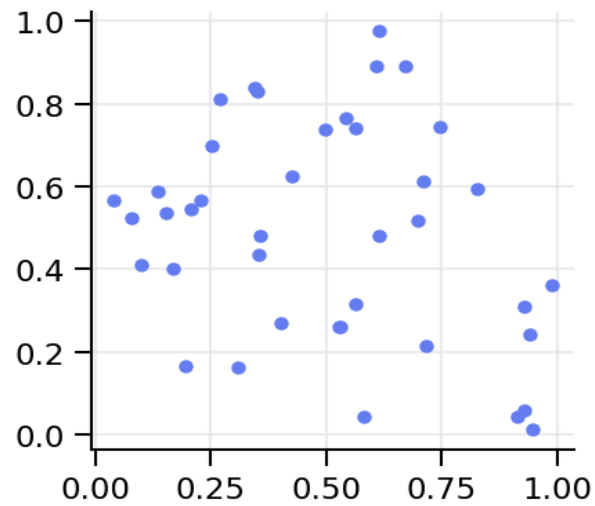
Hierarchical Agglomerative



Affinity Propagation

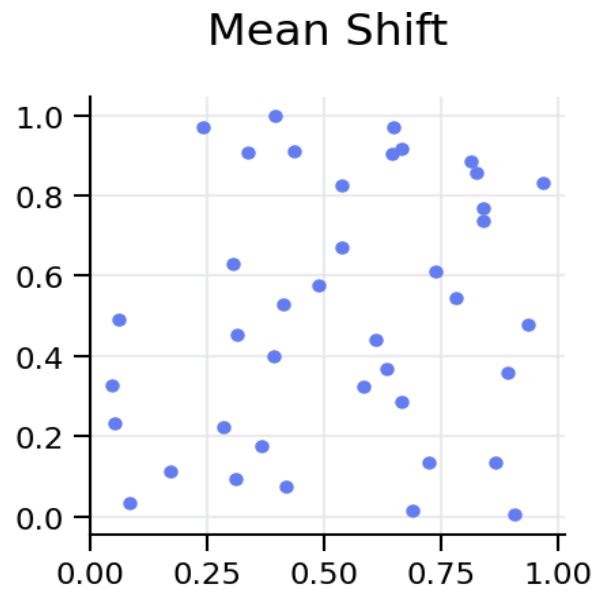
Affinity Propagation identifies representative data points called exemplars using message passing. It automatically decides the number of clusters but requires high memory.

Affinity Propagation



Mean Shift

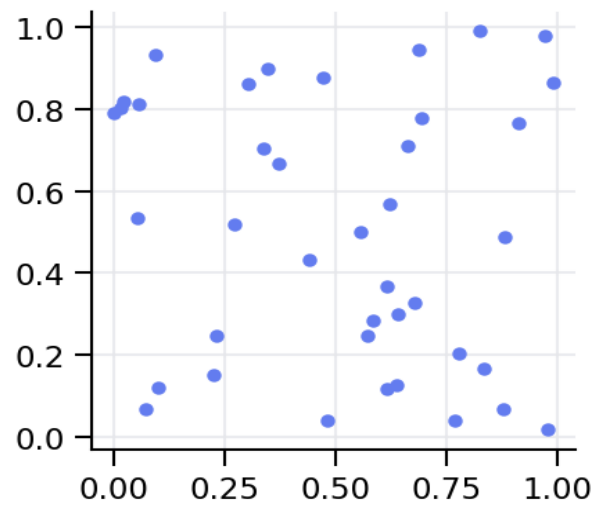
Mean Shift moves data points toward regions of maximum density. It does not require predefined clusters but is sensitive to bandwidth selection.



DBSCAN

DBSCAN groups points that are close to each other and marks isolated points as noise. It is effective for irregular cluster shapes and outlier detection.

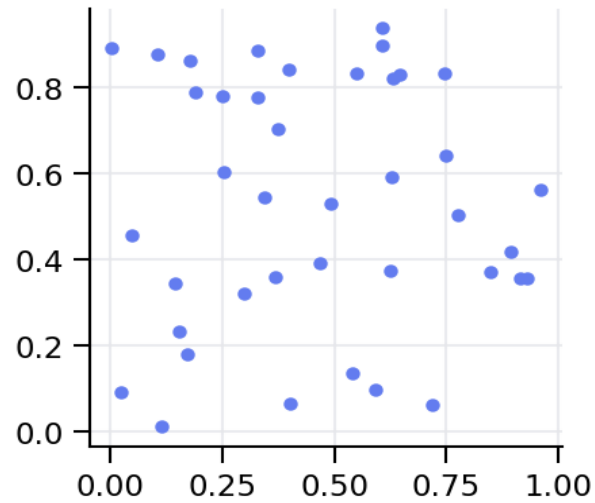
DBSCAN



OPTICS

OPTICS extends DBSCAN by handling clusters with different densities. It orders points to reveal cluster structure without a fixed epsilon.

OPTICS



BIRCH

BIRCH incrementally builds a tree-based structure for large datasets, making it fast and memory-efficient but less effective for complex cluster shapes.

